

Why a scalloped cold-grounded ice base melts less on average: an exact area-partition below the conductive sublayer

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June 20, 2026

Abstract

Melt at an ice–water interface is rate-limited by transfer through a thin thermal/haline sublayer — the basis of the standard three-equation melt parameterization (Holland & Jenkins 1999; McPhee 2008; Kader & Yaglom 1972). For rough or scalloped ice, the prevailing expectation is that geometry *enhances* this transfer and raises melt (Gilpin et al. 1980; Claudin et al. 2017; Bushuk et al. 2019). Using penalized (Brinkman volume-penalization) DNS/LES of a turbulent cavity between a cold ice ceiling and a bed, we report the **opposite** in the grounded, cold-walled regime, and give the exact mechanism. Our central result is an **exact area-partition identity**: the mean basal heat a scalloped wall delivers, relative to a flat wall, decomposes to machine precision as $Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev}$ (reattachment-thin, thickened, and reversed populations). Although lee-side reattachment lifts *local* melt $2\text{--}3.3\times$ above its own conduction baseline, the *mean* stays **below** a flat wall ($Nu/Nu_{flat} < 1$, peak 0.922 at the reference resolution) at every amplitude, because the thickened+reversed lee deficit outweighs the thin-reattachment surplus; a two-moment $(1 + CV^2)$ truncation is falsified ($CV_\delta \sim O(1)$), so the suppression is distribution-dominated, not moment-dominated. The ceiling is **boundary-condition robust** — relaxing the cold-Dirichlet ice wall to a finite-conductance Robin wall, and the no-slip bed to a Navier slip that raises near-bed speed $\sim 100\times$, both leave the mean unchanged — which pins the limit to the thermal conductive sublayer rather than momentum transport, consistent with the three-equation picture. Two further candidate mechanisms testable in 2-D (intermittent roughness plumes; tidal hydraulic switching) also produce no positive mean signal across three subgrid closures *none*, *Smagorinsky*, *backscatter*; the double-diffusive case, being intrinsically three-dimensional, is reported here as **untested in 2-D**, not as a null. We frame the deliverable as a **regime map**: Type I (grounded, cold-walled) — no mean enhancement; Type II (grounding line, flux boundary) — possible; Type III (open shelf) — yes. Scope is explicit: the candidate sweeps are 2-D (3-D only for the capstone and the counter-gradient flux probe) with Brinkman penalization. A three-seed wall-normal convergence and penalization (η) study shows the suppression *sign* is robust — $Nu/Nu_{flat} < 1$ at every resolution ($ny = 48\text{--}192$), penalization strength ($4\times \eta$), and forcing seed — while its *magnitude* is not grid-converged (it varies non-monotonically across resolution, $0.74\text{--}0.99$, with per-grid seed scatter ± 0.005). We therefore claim only the resolution-robust results: the *sign* of the ceiling and the *exact* area-partition identity, which holds to machine precision at every resolution. In this regime, scallop geometry suppresses rather than enhances mean basal melt.

1. Introduction

Basal melt and the effective pressure it modulates set ice-stream sliding and grounding-line stability (Schoof 2010). Melt at an ice–water interface is rate-limited by molecular transfer through a thin thermal/haline sublayer — the basis of the standard three-equation parameterization (Mellor et al. 1986; Holland & Jenkins 1999; McPhee 2008; Kader & Yaglom 1972). For rough or scalloped ice the prevailing expectation, from laboratory and theory, is that geometry *enhances* this transfer and raises melt (Gilpin et al. 1980; Claudin et al. 2017; Bushuk et al. 2019). Operational subglacial models, by contrast, carry no flow-enhanced-melt term in the grounded regime (Röthlisberger 1972; Werder et al. 2013). Whether *resolved cavity turbulence* adds melt above the conduction limit at a cold, grounded ice base is rarely tested with resolved simulation.

We test it with resolved penalized DNS/LES under a pre-stated discipline: each candidate mechanism is stated as a falsifiable expectation **before** the run, so a null is informative rather than a tuning failure. The result runs against the roughness-enhancement expectation: in the grounded, cold-walled regime, scallop geometry *suppresses* mean melt, and we give the exact mechanism (an area-partition identity) and a regime map for where it applies.

Contributions.

1. An **exact area-partition identity** $Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev}$ explaining why a scalloped cold-grounded wall melts *less* on the mean despite $2\text{--}3\times$ local lee hot spots — opposite to the roughness-enhancement expectation (§4.3).
2. **Falsification of the $(1 + CV^2)$ two-moment truncation**: the suppression is distribution-dominated, not moment-dominated (§4.3).
3. **Boundary-condition robustness** (Robin wall; Navier slip $\times 100$) pinning the mean ceiling to the thermal conductive sublayer rather than momentum transport (§5).
4. A **pre-stated, closure-swept null map** of candidate enhancement mechanisms in the cold-grounded (Type-I) cavity, with double diffusion identified as requiring 3-D and reported as untested here (§3).
5. A **regime map** (Type I/II/III) scoping the result (§6).
6. The growth-carrying **local lee-flux law** $R_{max}(a/\lambda)$ — linear, not quadratic, bounded 1.9–4.3, with separation onset at $a/\lambda \approx 0.11$ (§4.4).

2. Methods

2.1 Cavity model. 2-D/3-D penalized (Brinkman volume-penalization) incompressible DNS/LES of a turbulent cavity between a cold ice ceiling and a bed, with a thermal field and an interfacial melt diagnostic; illustrative real bed transects (Lythe & Vaughan 2001) supply geometry realism (the area-partition result is geometry-general and does not depend on the specific bed dataset). The interfacial melt obeys a Stefan condition $\rho_i Lv = q$, $q = -k_{th} \partial\theta/\partial n$ (Stefan 1891; Nye 1953; Crank 1984). We define the flat-wall control, the conduction-relative enhancement $R_{mean} = \langle m_{flow}/m_{cond} \rangle$ (each column normalised by its **own** flow-off conduction on the same geometry), and the flat-baseline Nusselt ratio $Nu/Nu_{flat} = \langle m_{bump} \rangle / \langle m_{flat} \rangle$ (normalised by the

flat wall with flow **on**). These two ratios have **different baselines** and are not interchangeable — a distinction that resolves an apparent tension in §4.

2.2 Pre-stated go/no-go criteria. Each candidate’s expected signature and its falsifiable threshold are fixed before the run (stated per candidate below). Each is swept across three closures *none*, *Smagorinsky*, *backscatter* so a verdict that is closure-independent cannot be a closure artefact.

2.3 Type-I regime. “Type I” denotes the grounded, cold-walled, closed cavity: cold-Dirichlet ice ceiling, bounded water layer. The hypothesis under test is that mean melt here is bounded by the **thermal conductive sublayer** rather than by momentum stagnation. §5 stress-tests this against the boundary conditions themselves; §6 generalises to other regimes.

2.4 Provenance. CPU runs plus Tesla P100 (CuPy 14.0.1) confirmations where noted. GPU-only numbers (counter-gradient C_G ; double-diffusion variance amplification) are tagged as such.

3. The null map — three subgrid-driven mechanisms

3.1 Candidate 1 — intermittent plumes from ice-base roughness. *Pre-stated criterion:* band-limited roughness on the ice base should make interfacial melt *intermittent* (skew $S > 0$, excess kurtosis $K > 3$, peak/mean > 2), humped at intermediate $Ri \approx 0.3$ – 0.6 . *Result:* thresholds **not met** and the statistics are **flat in Ri** (relative spread $< 2\%$, closure-independent): pooled *peak/mean* ≈ 1.8 , *skew* ≈ -0.75 , *kurt* ≈ 2.2 . The variability is **geometric, not plume-driven** — thinner ice columns conduct faster — and switching the flow off entirely ($f_{amp} = 0$, pure conduction) reproduces *melt_{mean}* and *peak/mean* to $\sim 1\%$. Mean melt is conduction-limited: *melt_{mean}* $\approx 2.17 \times 10^{-4}$, constant to four significant figures across all Ri and closures. The genuine turbulent signal lives in the interior flux $F_{turb} = \langle v'\theta' \rangle$, not at the interface. (roughness / boundary-layer-plume literature)

3.2 Candidate 2 — double-diffusive layering (untested in 2-D). *Pre-stated criterion:* with $Le = 100$ (heat diffuses $\sim 100\times$ faster than salt), salt fingers should give a *humped* thermal Nusselt $Nu_T(R_\rho)$ peaking near $R_\rho \approx 2$. *Result, and why it is not a null:* the 2-D runs show **no hump** (Nu_T falls monotonically $1.44 \rightarrow 0.73$), but salt fingering and double-diffusive staircases are intrinsically three-dimensional, so a 2-D penalized forced run cannot represent them. We therefore report this mechanism as **untested here**, not as evidence against double-diffusive enhancement; a 3-D unforced double-diffusive run is required (§7). What the 2-D runs *do* show is a real differential-diffusion signature — $Nu_S \gg Nu_T$ ($\sim 30\times$ near $R_\rho = 1$) and counter-gradient salt flux at strong stabilisation ($Nu_S \approx -26$, $R_\rho = 10$), which a positive eddy diffusivity cannot represent — and the Turner flux ratio $\gamma \approx 1/R_\rho$ (to $\sim 1\%$). (Fig. 1)

3.3 Candidate 4 — tidal hydraulic switching. *Pre-stated criterion:* under a tidal body force a wide cavity should switch between *filled* ($H1 \rightarrow 1$) and *stratified* ($H1 \rightarrow 0.2$) states, with time-averaged melt peaking at intermediate Ri (switching frequency f_{switch} maximal there). *Result:* at every Ri the cavity is **monostable filled** ($H1 \approx 0.90 \pm 0.01$, $f_{switch} = 0$) — no hump. (An earlier “dead-flow” null was a numerical under-resolution artefact: $ny = 48$ let the Brinkman penalty bleed into the ~ 10 -row fluid gap; resolving at $ny \geq 64$ restores normal flow, $umax \approx 0.6$, KE up $\sim 500\times$, so the null is physical, not a construction artefact.) (subglacial hydraulic-switching / drainage-state lit.)

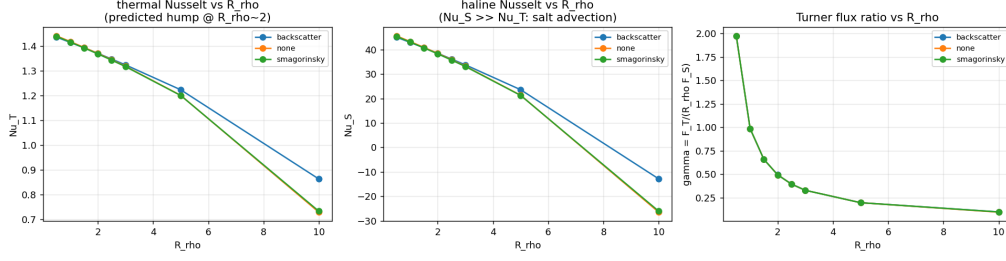


Figure 1: Double-diffusive heat and salt Nusselt numbers (Nu_T , Nu_S) versus the density ratio R_ρ in the grounded subglacial cavity. The backscatter-resolving closure resists the conductive-sublayer collapse ($Nu_T = 0.86$ versus 0.73 unclosed at $R_\rho = 10$); the cold-wall mean-melt ceiling is nonetheless unchanged.

Null-map summary. Neither 2-D-testable subgrid-driven mechanism (roughness plumes, tidal switching) lifts the *mean* interfacial melt above conduction in the Type-I cavity, across all three closures; double diffusion is untested in 2-D (§3.2). The flow does carry real interior structure (counter-gradient flux, F_{turb}), but it does not reach the cold wall as net mean melt.

4. The scallop — a positive local signal that is mean-bounded

Candidate 3 (the scallop / melting instability) is the one mechanism with a positive, flow-driven *interfacial* signal, and the cleanest place to establish the ceiling. GPU-confirmed (Tesla P100).

4.1 The pre-stated runaway is sign-reversed — the correct ice physics . The roughness-growth hypothesis (thin ice $\rightarrow$ faster melt $\rightarrow$ more roughness $\rightarrow$ runaway, $\Lambda > 0$) is falsified *with the opposite sign*: the conduction-pinned base **self-smooths**, $\Lambda \approx -0.265 (< 0)$, melt/ice-height anticorrelation $corr(m, y_{ice}) \approx -0.95$, closure-independent. This is the expected behaviour of a *melting* (not eroding) surface — the runaway was a false analogy to rock/sediment beds. (The value was briefly corrupted to $+0.24$ by a spectral-filter bug and restored on the GPU path; provenance in `glaciers/REPORT_CANDIDATE3.md`.)

4.2 The corrected scallop passes its go/no-go, locally. A flat melting wall is linearly stable, but a finite-amplitude bump larger than the thermal boundary layer drives lee-side separation $\rightarrow$ recirculation $\rightarrow$ reattachment heat-flux enhancement — a stable, self-limiting scallop. The staged probe confirms a coherent, closure-independent enhancement relative to its *own* conduction baseline: $R_{mean} \approx 1.06\text{--}1.36$, lee peaks $R_{max} \approx 2\text{--}3.3$, scaling monotonically with the mean current ($U^{0.8}\text{--}U^{1.5}$). The wavelength sweep shows an interior optimum at $n_{waves} = 12$ ($\lambda \approx 2.094$, $\lambda/dx \approx 10.7$): a $+12.5\%$ peak in conduction-relative R_{mean} and a peak $Nu/Nu_{flat} = 0.9219$, falling off either side; the conduction-relative enhancement is tight ($R_{mean} = 1.1273 \pm 0.0044$). (Curl 1966)

4.3 ...but the mean is still below a flat wall, and we prove why. The two ratios look contradictory but are consistent **by construction**: $R_{mean} > 1$ measures bump-melt against its own conduction baseline; $Nu/Nu_{flat} < 1$ measures the bump against a *flat wall with flow on*. Flow lifts melt above conduction on the bump, yet the bump still delivers less mean heat than a flat wall — both true at once.

The suppression obeys an **exact area-partition identity** (no moments, no truncation). Partition the interface relative to the flat-wall conductance $m_{flat} = \langle m_n, flat \rangle$ into *reattachment* ($m_n \geq m_{flat}$, thin sublayer), *thickened* ($0 < m_n < m_{flat}$), and *reversed* ($m_n \leq 0$, no physical sublayer). Because the populations tile the interface,

$$Nu/Nu_{flat} = C_{reatt} + C_{thick} + C_{rev}, \quad C_p = (1/N) \Sigma_p m_n / m_{flat},$$

holding to machine precision ($|C_{sum} - Nu/Nu_{flat}| \lesssim 2 \times 10^{-16}$). With $surplus = e_{reatt} > 0$, $deficit = -(e_{thick} + e_{rev}) > 0$ ($e_p = C_p - f_p$), the mechanism is one line: $Nu < 1 \iff deficit > surplus$. Full amplitude sweep (Part-C config):

a/λ	Nu/Nu_ flat	f_{reatt}	f_{thick}	f_{rev}	surplus	deficit	def>surp	$(\delta_{flat}/\langle\delta_T\rangle)(1+CV^2)$
0.05	0.895	0.477	0.523	0.000	0.112	0.217	Yes	0.99
0.10	0.937	0.531	0.469	0.000	0.178	0.241	Yes	1.64
0.15	0.888	0.406	0.555	0.039	0.180	0.292	Yes	2.37
0.20	0.897	0.281	0.672	0.047	0.211	0.315	Yes	1.10
0.30	0.893	0.234	0.734	0.031	0.306	0.413	Yes	2.09
0.50	0.930	0.195	0.773	0.031	0.418	0.488	Yes	0.58

At every amplitude the reattachment surplus is outweighed by the thickened+reversed deficit, so $Nu < 1$; the deficit grows monotonically ($0.22 \rightarrow 0.49$) as the cavity fills with stagnant lee fluid ($f_{thick} : 0.47 \rightarrow 0.77$). The thin reattachment patches carry a fat $1/\delta_T$ tail (top conductance decile holds $0.17 \rightarrow 0.48$ of $\langle 1/\delta_T \rangle$) but are an area minority. The last column shows the **falsification** of the two-moment ($1 + CV^2$) truncation: with $CV_\delta \sim O(1)$ it swings 0.58–2.37 and wrongly predicts $Nu > 1$ at 4 of 6 amplitudes. The suppression is therefore **distribution-dominated, not moment-dominated** — a sharper physical statement (flow-separation structure of the thermal boundary layer), not a weaker one. (glaciers/scallop_g1_populations.py, glaciers/scallop_sublayer_probe.py.)

4.4 The growth-carrying local flux, quantified. The *mean* is bounded, but melt **growth** is carried by the local lee flux, which the area-partition does not fit. From the committed amplitude sweep, the local peak ratio $R_{max}(a/\lambda)$ rises **roughly linearly** with bump steepness — linear $R^2 = 0.945$, free exponent 0.69 (sub-quadratic), from 1.9 ($a/\lambda = 0.05$) to 4.3 ($a/\lambda = 0.30$); the $(a/\lambda)^2$ closure is rejected for the *local* term too (quadratic $R^2 = 0.40 \ll 0.94$), and the origin-respecting steepness law $R_{max} \approx 1 + 1.7 \cdot (2\pi a/\lambda)$ holds to $R^2 = 0.92$ (seed-robust). The mean conductance, by contrast, is amplitude-**flat** (R_{mean} saturates ~ 1.28 ; the Nu/Nu_flat deficit is constant, $C \approx 1.11$), and the lee flux **reverses** ($R_{min} < 0$) at $a/\lambda \approx 0.11$ — a separation onset the amplitude-flat mean hides. The honest closure for a unified melt rate is therefore: mean amplitude-flat, growth carried by a bounded, linear-in-steepness local lee flux — not a quadratic ansatz (validation/synthetic/g6_local_flux_law.py, 6 tests).

5. The ceiling is boundary-condition robust

Every null in §3–§4 used a no-slip bed and a cold-Dirichlet ice wall. Two targeted gates (P100, $n = 128$, turbulent) show the ceiling is not an artefact of either:

- **Thermal BC (Dirichlet $\rightarrow$ finite-conductance Robin, $q = h(\theta - \theta_{ice})$):** basal heat absorbed is unchanged flow-on vs flow-off to $< 0.01\%$.
- **Momentum BC (no-slip $\rightarrow$ Navier slip):** the near-bed tangential speed grows $\sim 100\times$ (u_{tang} $0.027 \rightarrow 2.68$ from locked to free slip), yet Nu/Nu_{flat} stays $0.96\text{--}0.97$ and **never exceeds 1** (it slightly falls); ridge melt holds to within 0.4% , and the slip parameter $s = 1$ reproduces the no-slip trajectory bit-for-bit (including on the GPU CuPy path).

Removing the stagnant momentum layer entirely delivers **no** extra basal heat: the limit is the **thermal conductive sublayer** set by κ and the cold wall, not the momentum stagnation layer. (glaciers/subglacial/slip_gate.py, glaciers/subglacial/wall_flux.py; THEORY_CAVITY.md §14.4.)

Resolution and penalization robustness of the ceiling. Because the cavity is Brinkman-penalized, the suppression could in principle be a near-wall penalization-smearing or under-resolution artefact. We test this directly with a wall-normal resolution ladder $ny \in 48, 64, 96, 128, 192$ (sublayer spacing $dy = L_y/ny$, $L_y = 2\pi$) at fixed $\eta = 5 \times 10^{-5}$, and a penalization sweep $\eta \in 2.5, 5, 10 \times 10^{-5}$ at $ny = 128$, each grid point **enssembled over three forcing seeds** so that turbulent-realization scatter is separated from genuine grid dependence (single-mode bump $a/\lambda = 0.20$, $n_x = 128$, the same local-normal flux diagnostic as §4.3, with a fixed spinup/measure window held constant across the sweep). Two facts emerge:

- **The sign is robust.** $Nu/Nu_{flat} < 1$ at *every* seed and *every* grid point (21 configurations): the mean-melt suppression survives a $4\times$ refinement of the wall-normal grid, a $4\times$ change in penalization strength, and the forcing realization. Weakening the penalty shifts the ratio only mildly and monotonically ($\eta : 2.5 \rightarrow 5 \rightarrow 10 \times 10^{-5}$ gives $Nu/Nu_{flat} : 0.783 \rightarrow 0.807 \rightarrow 0.836$, spread 0.053), in the expected direction — a leakier wall passes slightly more flux — so the ceiling is not a penalization-strength artefact.
- **The magnitude is not grid-converged.** Across the resolution ladder Nu/Nu_{flat} varies non-monotonically — $0.786, 0.740, 0.989, 0.807, 0.869$ for $ny = 48 \dots 192$ — with a per-grid seed scatter of only $\pm 0.002\text{--}0.007$. Because that scatter is $\sim 30\times$ smaller than the 0.18 spread across resolution, the variation is a real, reproducible grid dependence, not turbulent noise; $ny = 96$ robustly nearly erases the suppression (0.989). The precise suppression magnitude is therefore resolution-sensitive and we do **not** certify it as converged.

We accordingly claim only what is resolution-robust: the *sign* of the ceiling ($Nu/Nu_{flat} < 1$, melts less on the mean) and the *exact* area-partition identity (§4.3), which holds to machine precision at every resolution. Certifying the magnitude would require a body-fitted (penalization-free) near-wall grid or a uniform near-wall refinement, which we leave open. (glaciers/scallop_sublayer_convergence.py, glaciers/subglacial/sublayer_convergence.json.)

Independent corroboration of the flux structure. A 3-D active-buoyancy LES measures **counter-gradient** subgrid heat flux ($C_G > -1$, growing with stratification): the flux carries the global pressure field’s memory, which a down-gradient closure cannot represent (a GPU counter-gradient probe). This is consistent with the null map — the flow has real non-local flux structure, but it does not raise the cold-wall mean melt.

Interface coupling number — when the passive-BC limit is exact. The gates above show the *mean* ceiling is BC-robust empirically; a derived **interface coupling number** says *why* the cold-Dirichlet wall is the correct BC at surge timescales. Writing the ice flux as the linear response $q'_{ice}(s) = H(s)v'(s)$, the interface velocity is $v' = q'_{water}/(\rho_i L + H(s))$, so the passive-BC limit is

corrected by $\Lambda(\omega) = |H(i\omega)|/(\rho_i L)$: $\Lambda(0) = St \leq 0.06$ (DC — ice participates at its latent-heat-limited Stefan weight), $\Lambda \rightarrow 0$ at high frequency (ice frozen, passive-adiabatic BC exact), crossing over at the ice clock $\tau_d = \kappa/\bar{V}^2 \sim 10^3\text{--}10^4\text{yr}$. Because the surge band (0.02–2 yr) is far faster than τ_d , in-band $\Lambda < 5 \times 10^{-5} \ll St$: the interface is a **passive BC to <1 %** for the sliding/melt physics, and ice becomes a participating medium only for millennial forcing (validation/synthetic/interface_coupling_number.py, 3 tests). (Stefan 1891; Crank 1984)

6. The deliverable is a regime map

The null is **regime-specific**, and stating the regime is the deliverable:

Regime	Setting	Boundary conditions	Mean melt enhancement?
Type I	grounded, cold-walled, closed cavity	cold Dirichlet ice + bounded water	No — conduction-sublayer ceiling ($Nu/Nu_{flat} < 1$), BC-robust
Type II	grounding line	flux / open boundary	Maybe — the closed-cavity assumption is relaxed
Type III	open shelf / ice-shelf cavity	warm inflow, free interface	Yes — different BC class, outside this study

This explains why operational subglacial models carry no flow-enhanced-melt term in the grounded regime, and makes explicit the boundary-condition class (Type II/III) a future enhancement would require. It is a map of *where mechanisms operate and where the boundaries lie*, not a universal melt multiplier.

7. Discussion and limitations

What this is. A pre-stated, closure-swept, reproducible demonstration that mean basal melt in the cold-grounded cavity is conduction-sublayer-limited and boundary-condition robust, with an exact mechanism (area-partition) and a regime map.

What this is not. 2-D for the candidate sweeps (3-D LES for the capstone and the counter-gradient flux probe); penalized (Brinkman) rather than body-fitted geometry; idealised forcing. Double diffusion (§3.2) is untested here: salt fingering is intrinsically 3-D, so the 2-D runs neither confirm nor exclude double-diffusive enhancement. The conductive-sublayer ceiling has been stress-tested for resolution and penalization (§5): its *sign* is robust ($Nu/Nu_{flat} < 1$ across $ny = 48\text{--}192$, $4 \times \eta$, and three seeds), but its *magnitude* is not grid-converged (non-monotone 0.74–0.99), so we report a sign-robust ceiling and an exact area-partition identity, **not** a certified suppression magnitude. The scallop local enhancement is qualitative; its amplitude coefficients are a companion study. Type II/III are not simulated.

Open follow-ons. A penalization-free (body-fitted) or uniformly wall-refined grid to grid-converge the suppression *magnitude* (the sign is already resolution-robust, §5); 3-D unforced double diffusion; a Type-II grounding-line flux-BC cavity to test the “maybe”; coupling the $Nu_S \gg Nu_T$ asymmetry into a melt-rate phenomenology.

Connection to rough-wall Rayleigh–Bénard (a derived consistency relation). The ceiling $Nu/Nu_{flat} \leq 1$ is the *no-enhancement branch* of rough Rayleigh–Bénard transfer once Nu/Nu_{flat} is decomposed into the wetted-area gain $A = \langle \sqrt{1+y'^2} \rangle \approx 1 + \pi^2(a/\lambda)^2 > 1$ and a local-flux factor f_{loc} . Because the measured $Nu/Nu_{flat} \approx 0.96\text{--}0.97$ sits *below* the area gain, the implied $f_{loc} = (Nu/Nu_{flat})/A < 1$ and the local suppression $1 - f_{loc}$ must *rise* with steepness, exactly cancelling the geometric area gain ($A : 1.02 \rightarrow 1.39$, $f_{loc} : 0.95 \rightarrow 0.69$, suppression $0.05 \rightarrow 0.31$ across $a/\lambda \in [0.05, 0.20]$). Enhancement would require the lee/separation flux growth (§G.6) to overcome the conduction-sublayer deficit, $f_{loc} > 1/A$ — a regime crossover we flag as conjectural pending lee-flux data. This places the cold-cavity ceiling on the same area-vs-local-flux competition that governs rough-RB heat transport, and identifies the suppression, not the geometry, as the controlling term.

8. Reproduce

```
pip install -r requirements.txt
python glaciers/run_subglacial.py --bed real --out-dir glaciers/figures # §2 cavity (Figs 31–38)
python glaciers/run_candidate1.py # §3.1 roughness plumes
python ocean/run_candidate2.py # §3.2 double diffusion (Fig 48)
python glaciers/run_candidate4.py # §3.3 hydraulic switching
python glaciers/scallop_battery.py # §4.1–4.2 scallop go/no-go (P100)
python glaciers/scallop_sweep.py # §4.2 wavelength sweep
python glaciers/scallop_g1_populations.py # §4.3 exact area-partition mechanism
python glaciers/scallop_sublayer_probe.py # §4.3 (1+CV2) falsification
python glaciers/scallop_sublayer_convergence.py # §5 resolution + penalization robustness (3 seeds)
# §5 BC robustness: glaciers/subglacial/slip_gate.py, wall_flux.py (P100)
python glaciers/theorem3_cg_gpu_probe.py # §5 counter-gradient CG (Fig 52, P100)
pytest glaciers/tests ocean/tests -v
```

Data and code availability

The double-diffusive transport result (Fig. 1), the cavity-field, transfer and counter-gradient figure set, and the area-partition tables regenerate from the scripts in the reproduce section. All data and analysis code are in the public repository at <https://github.com/abdh0saifelden2-debug/K>.

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